paul graham

on work that's yours

essays, volume one

contents

01	a project of one's own	june 2021	6
02	the bus ticket theory of genius	november 2019	12
03	how to think for yourself	november 2020	18
04	the lesson to unlearn	december 2019	26
05	how to work hard	june 2021	34
06	the right kind of stubborn	july 2024	40
07	what to do	march 2025	46
08	when to do what you love	september 2024	50
09	life is short	january 2016	54
10	how to do great work	july 2023	58

a few days ago, on the way home from school, my nine year old son told me he couldn't wait to get home to write more of the story he was working on. this made me as happy as anything i've heard him say — not just because he was excited about his story, but because he'd discovered this way of working. working on a project of your own is as different from ordinary work as skating is from walking. it's more fun, but also much more productive.

what proportion of great work has been done by people who were skating in this sense? if not all of it, certainly a lot.

there is something special about working on a project of your own. i wouldn't say exactly that you're happier. a better word would be excited, or engaged. you're happy when things are going well, but often they aren't. when i'm writing an essay, most of the time i'm worried and puzzled: worried that the essay will turn out badly, and puzzled because i'm groping for some idea that i can't see clearly enough. will i be able to pin it down with words? in the end i usually can, if i take long enough, but i'm never sure; the first few attempts often fail.

you have moments of happiness when things work out, but they don't last long, because then you're on to the next problem. so why do it at all? because to the kind of people who like working this way, nothing else feels as right. you feel as if you're an animal in its natural habitat, doing what you were meant to do — not always happy, maybe, but awake and alive.

many kids experience the excitement of working on projects of their own. the hard part is making this converge with the work you do as an adult. and our customs make it harder. we treat "playing" and "hobbies" as qualitatively different from "work". it's not clear to a kid building a treehouse that there's a direct (though long) route from that to architecture or engineering. and instead of pointing out the route, we conceal it, by implicitly treating the stuff kids do as different from real work. [1]

instead of telling kids that their treehouses could be on the path to the work they do as adults, we tell them the path goes through school. and unfortunately schoolwork tends to be very different from working on projects of one's own. it's usually neither a project, nor one's own. so as school gets more serious, working on projects of one's own is something that survives, if at all, as a thin thread off to the side.

it's a bit sad to think of all the high school kids turning their backs on building treehouses and sitting in class dutifully learning about darwin or newton to pass some exam, when the work that made darwin and newton famous was actually closer in spirit to building treehouses than studying for exams.

if i had to choose between my kids getting good grades and working on ambitious projects of their own, i'd pick the projects. and not because i'm an indulgent parent, but because i've been on the other end and i know which has more predictive value. when i was picking startups for y combinator, i didn't care about applicants' grades. but if they'd worked on projects of their own, i wanted to hear all about those. [2]

it may be inevitable that school is the way it is. i'm not saying we have to redesign it (though i'm not saying we don't), just that we should understand what it does to our attitudes to work — that it steers us toward the dutiful plodding kind of work, often using competition as bait, and away from skating.

there are occasionally times when schoolwork becomes a project of one's own. whenever i had to write a paper, that would become a project of my own — except in english classes, ironically, because the things one has to write in english classes are so bogus. and when i got to college and started taking cs classes, the programs i had to write became projects of my own. whenever i was writing or programming, i was usually skating, and that has been true ever since.

essness. writing code from scratch is also much more rewarding — so much more rewarding that a good programmer can end up net ahead, despite the shocking waste of characters. indeed, it may be one of the advantages of capitalism that it encourages such rewriting. a company that needs software to do something can't use the software already written to do it at another company, and thus has to write their own, which often turns out better. [5]

the natural alignment between skating and solving new problems is one of the reasons the payoffs from startups are so high. not only is the market price of unsolved problems higher, you also get a discount on productivity when you work on them. in fact, you get a double increase in productivity: when you're doing a clean-sheet design, it's easier to recruit skaters, and they get to spend all their time skating.

steve jobs knew a thing or two about skaters from having watched steve wozniak. if you can find the right people, you only have to tell them what to do at the highest level. they'll handle the details. indeed, they insist on it. for a project to feel like your own, you must have sufficient autonomy. you can't be working to order, or slowed down by bureaucracy.

one way to ensure autonomy is not to have a boss at all. there are two ways to do that: to be the boss yourself, and to work on projects outside of work. though they're at opposite ends of the scale financially, startups and open source projects have a lot in common, including the fact that they're often run by skaters. and indeed, there's a wormhole from one end of the scale to the other: one of the best ways to discover startup ideas is to work on a project just for fun.

if your projects are the kind that make money, it's easy to work on them. it's harder when they're not. and the hardest part, usually, is morale. that's where adults have it harder than kids. kids just plunge in and build their treehouse without worrying about whether they're wasting their time, or how it compares to other treehouses. and frankly we could learn a lot from kids here. the high standards most grownups have for "real" work do not always serve us well.

the most important phase in a project of one's own is at the beginning: when you go from thinking it might be cool to do x to actually doing x. and at that point high standards are not merely useless but positively harmful. there are a few people who start too many new projects, but far more, i suspect, who are deterred by fear of failure from starting projects that would have succeeded if they had.

but if we couldn't benefit as kids from the knowledge that our treehouses were on the path to grownup projects, we can at least benefit as grownups from knowing that our projects are on a path that stretches back to treehouses. remember that careless confidence you had as a kid when starting something new? that would be a powerful thing to recapture.

if it's harder as adults to retain that kind of confidence, we at least tend to be more aware of what we're doing. kids bounce, or are herded, from one kind of work to the next, barely realizing what's happening to them. whereas we know more about different types of work and have more control over which we do. ideally we can have the best of both worlds: to be deliberate in choosing to work on projects of our own, and carelessly confident in starting new ones.

---

notes

1 "hobby" is a curious word. now it means work that isn't *real* work — work that one is not to be judged by — but originally it just meant an obsession in a fairly general sense (even a political opinion, for example) that

everyone knows that to do great work you need both natural ability and determination. but there's a third ingredient that's not as well understood: an obsessive interest in a particular topic.

to explain this point i need to burn my reputation with some group of people, and i'm going to choose bus ticket collectors. there are people who collect old bus tickets. like many collectors, they have an obsessive interest in the minutiae of what they collect. they can keep track of distinctions between different types of bus tickets that would be hard for the rest of us to remember. because we don't care enough. what's the point of spending so much time thinking about old bus tickets?

which leads us to the second feature of this kind of obsession: there is no point. a bus ticket collector's love is disinterested. they're not doing it to impress us or to make themselves rich, but for its own sake.

when you look at the lives of people who've done great work, you see a consistent pattern. they often begin with a bus ticket collector's obsessive interest in something that would have seemed pointless to most of their contemporaries. one of the most striking features of darwin's book about his voyage on the beagle is the sheer depth of his interest in natural history. his curiosity seems infinite. ditto for ramanujan, sitting by the hour working out on his slate what happens to series.

it's a mistake to think they were "laying the groundwork" for the discoveries they made later. there's too much intention in that metaphor. like bus ticket collectors, they were doing it because they liked it.

but there is a difference between ramanujan and a bus ticket collector. series matter, and bus tickets don't.

if i had to put the recipe for genius into one sentence, that might be it: to have a disinterested obsession with something that matters.

aren't i forgetting about the other two ingredients? less than you might think. an obsessive interest in a topic is both a proxy for ability and a substitute for determination. unless you have sufficient mathematical aptitude, you won't find series interesting. and when you're obsessively interested in something, you don't need as much determination: you don't need to push yourself as hard when curiosity is pulling you.

an obsessive interest will even bring you luck, to the extent anything can. chance, as pasteur said, favors the prepared mind, and if there's one thing an obsessed mind is, it's prepared.

the disinterestedness of this kind of obsession is its most important feature. not just because it's a filter for earnestness, but because it helps you discover new ideas.

the paths that lead to new ideas tend to look unpromising. if they looked promising, other people would already have explored them. how do the people who do great work discover these paths that others overlook? the popular story is that they simply have better vision: because they're so talented, they see paths that others miss. but if you look at the way great discoveries are made, that's not what happens. darwin didn't pay closer attention to individual species than other people because he saw that this would lead to great discoveries, and they didn't. he was just really, really interested in such things.

darwin couldn't turn it off. neither could ramanujan. they didn't discover the hidden paths that they did because they seemed promising, but because they couldn't help it. that's what allowed them to follow paths that someone who was merely ambitious would have ignored.

what rational person would decide that the way to write great novels was to begin by spending several years creating an imaginary elvish language, like tolkien, or visiting every household in southwestern britain, like trollope? no one, including tolkien and trollope.

one interesting thing about the bus ticket theory is that it may help explain why different types of people excel at different kinds of work. interest is much more unevenly distributed than ability. if natural ability is all you need to do great work, and natural ability is evenly distributed, you have to invent elaborate theories to explain the skewed distributions we see among those who actually do great work in various fields. but it may be that much of the skew has a simpler explanation: different people are interested in different things.

the bus ticket theory also explains why people are less likely to do great work after they have children. here interest has to compete not just with external obstacles, but with another interest, and one that for most people is extremely powerful. it's harder to find time for work after you have kids, but that's the easy part. the real change is that you don't want to.

but the most exciting implication of the bus ticket theory is that it suggests ways to encourage great work. if the recipe for genius is simply natural ability plus hard work, all we can do is hope we have a lot of ability, and work as hard as we can. but if interest is a critical ingredient in genius, we may be able, by cultivating interest, to cultivate genius.

for example, for the very ambitious, the bus ticket theory suggests that the way to do great work is to relax a little. instead of gritting your teeth and diligently pursuing what all your peers agree is the most promising line of research, maybe you should try doing something just for fun. and if you're stuck, that may be the vector along which to break out.

i've always liked hamming's famous double-barrelled question: what are the most important problems in your field, and why aren't you working on one of them? it's a great way to shake yourself up. but it may be overfitting a bit. it might be at least as useful to ask yourself: if you could take a year off to work on something that probably wouldn't be important but would be really interesting, what would it be?

the bus ticket theory also suggests a way to avoid slowing down as you get older. perhaps the reason people have fewer new ideas as they get older is not simply that they're losing their edge. it may also be because once you become established, you can no longer mess about with irresponsible side projects the way you could when you were young and no one cared what you did.

the solution to that is obvious: remain irresponsible. it will be hard, though, because the apparently random projects you take up to stave off decline will read to outsiders as evidence of it. and you yourself won't know for sure that they're wrong. but it will at least be more fun to work on what you want.

it may even be that we can cultivate a habit of intellectual bus ticket collecting in kids. the usual plan in education is to start with a broad, shallow focus, then gradually become more specialized. but i've done the opposite with my kids. i know i can count on their school to handle the broad, shallow part, so i take them deep.

when they get interested in something, however random, i encourage them to go preposterously, bus ticket collectorly, deep. i don't do this because of the bus ticket theory. i do it because i want them to feel the joy of learning, and they're never going to feel that about something i'm making them learn. it has to be something they're interested in. i'm just following the path of least resistance; depth is a byproduct. but if in trying to show them the joy of learning i also end up training them to go deep, so much the better.

will it have any effect? i have no idea. but that uncertainty may be the most interesting point of all. there is so much more to learn about how to do great work. as old as human civilization feels, it's reall-

> something they wouldn't otherwise be. not in a way that sticks. the most you can do on behalf of a topic is to make sure it gets a fair showing — to make it clear to them, for example, that there's more to math than the dull drills they do in school. after that it's up to the child.

thanks to marc andreessen, trevor blackwell, patrick collison, kevin lacker, jessica livingston, jackie mcdonough, robert morris, lisa randall, zak stone, and my 7 year old for reading drafts of this.

there are some kinds of work that you can't do well without thinking differently from your peers. to be a successful scientist, for example, it's not enough just to be correct. your ideas have to be both correct and novel. you can't publish papers saying things other people already know. you need to say things no one else has realized yet.

the same is true for investors. it's not enough for a public market investor to predict correctly how a company will do. if a lot of other people make the same prediction, the stock price will already reflect it, and there's no room to make money. the only valuable insights are the ones most other investors don't share.

you see this pattern with startup founders too. you don't want to start a startup to do something that everyone agrees is a good idea, or there will already be other companies doing it. you have to do something that sounds to most other people like a bad idea, but that you know isn't — like writing software for a tiny computer used by a few thousand hobbyists, or starting a site to let people rent airbeds on strangers' floors.

ditto for essayists. an essay that told people things they already knew would be boring. you have to tell them something new.

but this pattern isn't universal. in fact, it doesn't hold for most kinds of work. in most kinds of work — to be an administrator, for example — all you need is the first half. all you need is to be right. it's not essential that everyone else be wrong.

there's room for a little novelty in most kinds of work, but in practice there's a fairly sharp distinction between the kinds of work where it's essential to be independent-minded, and the kinds where it's not.

i wish someone had told me about this distinction when i was a kid, because it's one of the most important things to think about when you're deciding what kind of work you want to do. do you want to do the kind of work where you can only win by thinking differently from everyone else? i suspect most people's unconscious mind will answer that question before their conscious mind has a chance to. i know mine does.

independent-mindedness seems to be more a matter of nature than nurture. which means if you pick the wrong type of work, you're going to be unhappy. if you're naturally independent-minded, you're going to find it frustrating to be a middle manager. and if you're naturally conventional-minded, you're going to be sailing into a headwind if you try to do original research.

one difficulty here, though, is that people are often mistaken about where they fall on the spectrum from conventional- to independent-minded. conventional-minded people don't like to think of themselves as conventional-minded. and in any case, it genuinely feels to them as if they make up their own minds about everything. it's just a coincidence that their beliefs are identical to their peers'. and the independent-minded, meanwhile, are often unaware how different their ideas are from conventional ones, at least till they state them publicly. [1]

by the time they reach adulthood, most people know roughly how smart they are (in the narrow sense of ability to solve pre-set problems), because they're constantly being tested and ranked according to it. but schools generally ignore independent-mindedness, except to the extent they try to suppress it. so we don't get anything like the same kind of feedback about how independent-minded we are.

there may even be a phenomenon like dunning-kruger at work, where the most conventional-minded people are confident that they're independent-minded, while the genuinely independent-minded worry they might not be independent-minded enough.

you can expand the source of influences in time as well as space, by reading history. when i read history i do it not just to learn what happened, but to try to get inside the heads of people who lived in the past. how did things look to them? this is hard to do, but worth the effort for the same reason it's worth travelling far to triangulate a point.

you can also take more explicit measures to prevent yourself from automatically adopting conventional opinions. the most general is to cultivate an attitude of skepticism. when you hear someone say something, stop and ask yourself "is that true?" don't say it out loud. i'm not suggesting that you impose on everyone who talks to you the burden of proving what they say, but rather that you take upon yourself the burden of evaluating what they say.

treat it as a puzzle. you know that some accepted ideas will later turn out to be wrong. see if you can guess which. the end goal is not to find flaws in the things you're told, but to find the new ideas that had been concealed by the broken ones. so this game should be an exciting quest for novelty, not a boring protocol for intellectual hygiene. and you'll be surprised, when you start asking "is this true?", how often the answer is not an immediate yes. if you have any imagination, you're more likely to have too many leads to follow than too few.

more generally your goal should be not to let anything into your head unexamined, and things don't always enter your head in the form of statements. some of the most powerful influences are implicit. how do you even notice these? by standing back and watching how other people get their ideas.

when you stand back at a sufficient distance, you can see ideas spreading through groups of people like waves. the most obvious are in fashion: you notice a few people wearing a certain kind of shirt, and then more and more, until half the people around you are wearing the same shirt. you may not care much what you wear, but there are intellectual fashions too, and you definitely don't want to participate in those. not just because you want sovereignty over your own thoughts, but because unfashionable ideas are disproportionately likely to lead somewhere interesting. the best place to find undiscovered ideas is where no one else is looking. [4]

---

to go beyond this general advice, we need to look at the internal structure of independent-mindedness — at the individual muscles we need to exercise, as it were. it seems to me that it has three components: fastidiousness about truth, resistance to being told what to think, and curiosity.

fastidiousness about truth means more than just not believing things that are false. it means being careful about degree of belief. for most people, degree of belief rushes unexamined toward the extremes: the unlikely becomes impossible, and the probable becomes certain. [5] to the independent-minded, this seems unpardonably sloppy. they're willing to have anything in their heads, from highly speculative hypotheses to (apparent) tautologies, but on subjects they care about, everything has to be labelled with a carefully considered degree of belief. [6]

the independent-minded thus have a horror of ideologies, which require one to accept a whole collection of beliefs at once, and to treat them as articles of faith. to an independent-minded person that would seem revolting, just as it would seem to someone fastidious about food to take a bite of a submarine sandwich filled with a large variety of ingredients of indeterminate age and provenance.

without this fastidiousness about truth, you can't be truly independent-minded. it's not enough just to have resistance to being told what to think. those kind of people reject conventional ideas only to replace them with the most random conspiracy theories. and since these conspiracy theories have often

dness. some independent-minded people are openly subversive, and others are quietly curious. they all know the secret handshake though.

is there a way to cultivate curiosity? to start with, you want to avoid situations that suppress it. how much does the work you're currently doing engage your curiosity? if the answer is "not much," maybe you should change something.

the most important active step you can take to cultivate your curiosity is probably to seek out the topics that engage it. few adults are equally curious about everything, and it doesn't seem as if you can choose which topics interest you. so it's up to you to find them. or invent them, if necessary.

another way to increase your curiosity is to indulge it, by investigating things you're interested in. curiosity is unlike most other appetites in this respect: indulging it tends to increase rather than to sate it. questions lead to more questions.

curiosity seems to be more individual than fastidiousness about truth or resistance to being told what to think. to the degree people have the latter two, they're usually pretty general, whereas different people can be curious about very different things. so perhaps curiosity is the compass here. perhaps, if your goal is to discover novel ideas, your motto should not be "do what you love" so much as "do what you're curious about."

---

notes

1 one convenient consequence of the fact that no one identifies as conventional-minded is that you can say what you like about conventional-minded people without getting in too much trouble. when i wrote "the four quadrants of conformism" i expected a firestorm of rage from the aggressively conventional-minded, but in fact it was quite muted. they sensed that there was something about the essay that they disliked intensely, but they had a hard time finding a specific passage to pin it on.

2 when i ask myself what in my life is like high school, the answer is twitter. it's not just full of conventional-minded people, as anything its size will inevitably be, but subject to violent storms of conventional-mindedness that remind me of descriptions of jupiter. but while it probably is a net loss to spend time there, it has at least made me think more about the distinction between independent- and conventional-mindedness, which i probably wouldn't have done otherwise.

3 the decrease in independent-mindedness in growing startups is still an open problem, but there may be solutions.

founders can delay the problem by making a conscious effort only to hire independent-minded people. which of course also has the ancillary benefit that they have better ideas.

another possible solution is to create policies that somehow disrupt the force of conformism, much as control rods slow chain reactions, so that the conventional-minded aren't as dangerous. the physical separation of lockheed's skunk works may have had this as a side benefit. recent examples suggest employee forums like slack may not be an unmitigated good.

the most radical solution would be to grow revenues without growing the company. you think hiring that junior pr person will be cheap, compared to a programmer, but what will be the effect on the average level of independent-mindedness in your company? (the growth in staff relative to faculty seems to have had a similar effect on universities.) perhaps the rule about outsourcing work that's not your "core

the most damaging thing you learned in school wasn't something you learned in any specific class. it was learning to get good grades.

when i was in college, a particularly earnest philosophy grad student once told me that he never cared what grade he got in a class, only what he learned in it. this stuck in my mind because it was the only time i ever heard anyone say such a thing.

for me, as for most students, the measurement of what i was learning completely dominated actual learning in college. i was fairly earnest; i was genuinely interested in most of the classes i took, and i worked hard. and yet i worked by far the hardest when i was studying for a test.

in theory, tests are merely what their name implies: tests of what you've learned in the class. in theory you shouldn't have to prepare for a test in a class any more than you have to prepare for a blood test. in theory you learn from taking the class, from going to the lectures and doing the reading and/or assignments, and the test that comes afterward merely measures how well you learned.

in practice, as almost everyone reading this will know, things are so different that hearing this explanation of how classes and tests are meant to work is like hearing the etymology of a word whose meaning has changed completely. in practice, the phrase "studying for a test" was almost redundant, because that was when one really studied. the difference between diligent and slack students was that the former studied hard for tests and the latter didn't. no one was pulling all-nighters two weeks into the semester.

even though i was a diligent student, almost all the work i did in school was aimed at getting a good grade on something.

to many people, it would seem strange that the preceding sentence has a "though" in it. aren't i merely stating a tautology? isn't that what a diligent student is, a straight-a student? that's how deeply the conflation of learning with grades has infused our culture.

is it so bad if learning is conflated with grades? yes, it is bad. and it wasn't till decades after college, when i was running y combinator, that i realized how bad it is.

i knew of course when i was a student that studying for a test is far from identical with actual learning. at the very least, you don't retain knowledge you cram into your head the night before an exam. but the problem is worse than that. the real problem is that most tests don't come close to measuring what they're supposed to.

if tests truly were tests of learning, things wouldn't be so bad. getting good grades and learning would converge, just a little late. the problem is that nearly all tests given to students are terribly hackable. most people who've gotten good grades know this, and know it so well they've ceased even to question it. you'll see when you realize how naive it sounds to act otherwise.

suppose you're taking a class on medieval history and the final exam is coming up. the final exam is supposed to be a test of your knowledge of medieval history, right? so if you have a couple days between now and the exam, surely the best way to spend the time, if you want to do well on the exam, is to read the best books you can find about medieval history. then you'll know a lot about it, and do well on the exam.

no, no, no, experienced students are saying to themselves. if you merely read good books on medieval history, most of the stuff you learned wouldn't be on the test. it's not good books you want to read, but the lecture notes and assigned reading in this class. and even most of that you can ignore, because you only have to worry about the sort of thing that could turn up as a test question. you're looking for

you don't care about that are mostly memorization, the random "extracurricular activities" you have to participate in to show you're "well-rounded," the standardized tests as artificial as chess, the "essay" you have to write that's presumably meant to hit some very specific target, but you're not told what.

as well as being bad in what it does to kids, this test is also bad in the sense of being very hackable. so hackable that whole industries have grown up to hack it. this is the explicit purpose of test-prep companies and admissions counsellors, but it's also a significant part of the function of private schools.

why is this particular test so hackable? i think because of what it's measuring. although the popular story is that the way to get into a good college is to be really smart, admissions officers at elite colleges neither are, nor claim to be, looking only for that. what are they looking for? they're looking for people who are not simply smart, but admirable in some more general sense. and how is this more general admirableness measured? the admissions officers feel it. in other words, they accept who they like.

so what college admissions is a test of is whether you suit the taste of some group of people. well, of course a test like that is going to be hackable. and because it's both very hackable and there's (thought to be) a lot at stake, it's hacked like nothing else. that's why it distorts your life so much for so long.

it's no wonder high school students often feel alienated. the shape of their lives is completely artificial.

but wasting your time is not the worst thing the educational system does to you. the worst thing it does is to train you that the way to win is by hacking bad tests. this is a much subtler problem that i didn't recognize until i saw it happening to other people.

when i started advising startup founders at y combinator, especially young ones, i was puzzled by the way they always seemed to make things overcomplicated. how, they would ask, do you raise money? what's the trick for making venture capitalists want to invest in you? the best way to make vcs want to invest in you, i would explain, is to actually be a good investment. even if you could trick vcs into investing in a bad startup, you'd be tricking yourselves too. you're investing time in the same company you're asking them to invest money in. if it's not a good investment, why are you even doing it?

oh, they'd say, and then after a pause to digest this revelation, they'd ask: what makes a startup a good investment?

so i would explain that what makes a startup promising, not just in the eyes of investors but in fact, is growth. ideally in revenue, but failing that in usage. what they needed to do was get lots of users.

how does one get lots of users? they had all kinds of ideas about that. they needed to do a big launch that would get them "exposure." they needed influential people to talk about them. they even knew they needed to launch on a tuesday, because that's when one gets the most attention.

no, i would explain, that is not how to get lots of users. the way you get lots of users is to make the product really great. then people will not only use it but recommend it to their friends, so your growth will be exponential once you get it started.

at this point i've told the founders something you'd think would be completely obvious: that they should make a good company by making a good product. and yet their reaction would be something like the reaction many physicists must have had when they first heard about the theory of relativity: a mixture of astonishment at its apparent genius, combined with a suspicion that anything so weird couldn't possibly be right. ok, they would say, dutifully. and could you introduce us to such-and-such influential person? and remember, we want to launch on tuesday.

like a startup? i'm not going to chase down all the implications now. what i want to focus on here is what it means for individuals.

to start with, it means that most ambitious kids graduating from college have something they may want to unlearn. but it also changes how you look at the world. instead of looking at all the different kinds of work people do and thinking of them vaguely as more or less appealing, you can now ask a very specific question that will sort them in an interesting way: to what extent do you win at this kind of work by hacking bad tests?

it would help if there was a way to recognize bad tests quickly. is there a pattern here? it turns out there is.

tests can be divided into two kinds: those that are imposed by authorities, and those that aren't. tests that aren't imposed by authorities are inherently unhackable, in the sense that no one is claiming they're tests of anything more than they actually test. a football match, for example, is simply a test of who wins, not which team is better. you can tell that from the fact that commentators sometimes say afterward that the better team won. whereas tests imposed by authorities are usually proxies for something else. a test in a class is supposed to measure not just how well you did on that particular test, but how much you learned in the class. while tests that aren't imposed by authorities are inherently unhackable, those imposed by authorities have to be made unhackable. usually they aren't. so as a first approximation, bad tests are roughly equivalent to tests imposed by authorities.

you might actually like to win by hacking bad tests. presumably some people do. but i bet most people who find themselves doing this kind of work don't like it. they just take it for granted that this is how the world works, unless you want to drop out and be some kind of hippie artisan.

i suspect many people implicitly assume that working in a field with bad tests is the price of making lots of money. but that, i can tell you, is false. it used to be true. in the mid-twentieth century, when the economy was composed of oligopolies, the only way to the top was by playing their game. but it's not true now. there are now ways to get rich by doing good work, and that's part of the reason people are so much more excited about getting rich than they used to be. when i was a kid, you could either become an engineer and make cool things, or make lots of money by becoming an "executive." now you can make lots of money by making cool things.

hacking bad tests is becoming less important as the link between work and authority erodes. the erosion of that link is one of the most important trends happening now, and we see its effects in almost every kind of work people do. startups are one of the most visible examples, but we see much the same thing in writing. writers no longer have to submit to publishers and editors to reach readers; now they can go direct.

the more i think about this question, the more optimistic i get. this seems one of those situations where we don't realize how much something was holding us back until it's eliminated. and i can foresee the whole bogus edifice crumbling. imagine what happens as more and more people start to ask themselves if they want to win by hacking bad tests, and decide that they don't. the kinds of work where you win by hacking bad tests will be starved of talent, and the kinds where you win by doing good work will see an influx of the most ambitious people. and as hacking bad tests shrinks in importance, education will evolve to stop training us to do it. imagine what the world could look like if that happened.

this is not just a lesson for individuals to unlearn, but one for society to unlearn, and we'll be amazed at the energy that's liberated when we do.

it might not seem there's much to learn about how to work hard. anyone who's been to school knows what it entails, even if they chose not to do it. there are 12 year olds who work amazingly hard. and yet when i ask if i know more about working hard now than when i was in school, the answer is definitely yes.

one thing i know is that if you want to do great things, you'll have to work very hard. i wasn't sure of that as a kid. schoolwork varied in difficulty; one didn't always have to work super hard to do well. and some of the things famous adults did, they seemed to do almost effortlessly. was there, perhaps, some way to evade hard work through sheer brilliance? now i know the answer to that question. there isn't.

the reason some subjects seemed easy was that my school had low standards. and the reason famous adults seemed to do things effortlessly was years of practice; they made it look easy.

of course, those famous adults usually had a lot of natural ability too. there are three ingredients in great work: natural ability, practice, and effort. you can do pretty well with just two, but to do the best work you need all three: you need great natural ability *and* to have practiced a lot *and* to be trying very hard. [1]

bill gates, for example, was among the smartest people in business in his era, but he was also among the hardest working. "i never took a day off in my twenties," he said. "not one." it was similar with lionel messi. he had great natural ability, but when his youth coaches talk about him, what they remember is not his talent but his dedication and his desire to win. p. g. wodehouse would probably get my vote for best english writer of the 20th century, if i had to choose. certainly no one ever made it look easier. but no one ever worked harder. at 74, he wrote

> with each new book of mine i have, as i say, the feeling that this time i have picked a lemon in the garden of literature. a good thing, really, i suppose. keeps one up on one's toes and makes one rewrite every sentence ten times. or in many cases twenty times.

sounds a bit extreme, you think. and yet bill gates sounds even more extreme. not one day off in ten years? these two had about as much natural ability as anyone could have, and yet they also worked about as hard as anyone could work. you need both.

that seems so obvious, and yet in practice we find it slightly hard to grasp. there's a faint xor between talent and hard work. it comes partly from popular culture, where it seems to run very deep, and partly from the fact that the outliers are so rare. if great talent and great drive are both rare, then people with both are rare squared. most people you meet who have a lot of one will have less of the other. but you'll need both if you want to be an outlier yourself. and since you can't really change how much natural talent you have, in practice doing great work, insofar as you can, reduces to working very hard.

it's straightforward to work hard if you have clearly defined, externally imposed goals, as you do in school. there is some technique to it: you have to learn not to lie to yourself, not to procrastinate (which is a form of lying to yourself), not to get distracted, and not to give up when things go wrong. but this level of discipline seems to be within the reach of quite young children, if they want it.

what i've learned since i was a kid is how to work toward goals that are neither clearly defined nor externally imposed. you'll probably have to learn both if you want to do really great things.

the most basic level of which is simply to feel you should be working without anyone telling you to. now, when i'm not working hard, alarm bells go off. i can't be sure i'm getting anywhere when i'm working hard, but i can be sure i'm getting nowhere when i'm not, and it feels awful. [2]

the only way to find the limit is by crossing it. cultivate a sensitivity to the quality of the work you're doing, and then you'll notice if it decreases because you're working too hard. honesty is critical here, in both directions: you have to notice when you're being lazy, but also when you're working too hard. and if you think there's something admirable about working too hard, get that idea out of your head. you're not merely getting worse results, but getting them because you're showing off — if not to other people, then to yourself. [6]

finding the limit of working hard is a constant, ongoing process, not something you do just once. both the difficulty of the work and your ability to do it can vary hour to hour, so you need to be constantly judging both how hard you're trying and how well you're doing.

trying hard doesn't mean constantly pushing yourself to work, though. there may be some people who do, but i think my experience is fairly typical, and i only have to push myself occasionally when i'm starting a project or when i encounter some sort of check. that's when i'm in danger of procrastinating. but once i get rolling, i tend to keep going.

what keeps me going depends on the type of work. when i was working on viaweb, i was driven by fear of failure. i barely procrastinated at all then, because there was always something that needed doing, and if i could put more distance between me and the pursuing beast by doing it, why wait? [7] whereas what drives me now, writing essays, is the flaws in them. between essays i fuss for a few days, like a dog circling while it decides exactly where to lie down. but once i get started on one, i don't have to push myself to work, because there's always some error or omission already pushing me.

i do make some amount of effort to focus on important topics. many problems have a hard core at the center, surrounded by easier stuff at the edges. working hard means aiming toward the center to the extent you can. some days you may not be able to; some days you'll only be able to work on the easier, peripheral stuff. but you should always be aiming as close to the center as you can without stalling.

the bigger question of what to do with your life is one of these problems with a hard core. there are important problems at the center, which tend to be hard, and less important, easier ones at the edges. so as well as the small, daily adjustments involved in working on a specific problem, you'll occasionally have to make big, lifetime-scale adjustments about which type of work to do. and the rule is the same: working hard means aiming toward the center — toward the most ambitious problems.

by center, though, i mean the actual center, not merely the current consensus about the center. the consensus about which problems are most important is often mistaken, both in general and within specific fields. if you disagree with it, and you're right, that could represent a valuable opportunity to do something new.

the more ambitious types of work will usually be harder, but although you should not be in denial about this, neither should you treat difficulty as an infallible guide in deciding what to do. if you discover some ambitious type of work that's a bargain in the sense of being easier for you than other people, either because of the abilities you happen to have, or because of some new way you've found to approach it, or simply because you're more excited about it, by all means work on that. some of the best work is done by people who find an easy way to do something hard.

as well as learning the shape of real work, you need to figure out which kind you're suited for. and that doesn't just mean figuring out which kind your natural abilities match the best; it doesn't mean that if you're 7 feet tall, you have to play basketball. what you're suited for depends not just on your talents but perhaps even more on your interests. a deep interest in a topic makes people work harder than any amount of discipline can.

notes

1 in "the bus ticket theory of genius" i said the three ingredients in great work were natural ability, determination, and interest. that's the formula in the preceding stage; determination and interest yield practice and effort.

2 i mean this at a resolution of days, not hours. you'll often get somewhere while not working in the sense that the solution to a problem comes to you while taking a shower, or even in your sleep, but only because you were working hard on it the day before.

it's good to go on vacation occasionally, but when i go on vacation, i like to learn new things. i wouldn't like just sitting on a beach.

3 the thing kids do in school that's most like the real version is sports. admittedly because many sports originated as games played in schools. but in this one area, at least, kids are doing exactly what adults do.

in the average american high school, you have a choice of pretending to do something serious, or seriously doing something pretend. arguably the latter is no worse.

4 knowing what you want to work on doesn't mean you'll be able to. most people have to spend a lot of their time working on things they don't want to, especially early on. but if you know what you want to do, you at least know what direction to nudge your life in.

5 the lower time limits for intense work suggest a solution to the problem of having less time to work after you have kids: switch to harder problems. in effect i did that, though not deliberately.

6 some cultures have a tradition of performative hard work. i don't love this idea, because (a) it makes a parody of something important and (b) it causes people to wear themselves out doing things that don't matter. i don't know enough to say for sure whether it's net good or bad, but my guess is bad.

7 one of the reasons people work so hard on startups is that startups can fail, and when they do, that failure tends to be both decisive and conspicuous.

8 it's ok to work on something to make a lot of money. you need to solve the money problem somehow, and there's nothing wrong with doing that efficiently by trying to make a lot at once. i suppose it would even be ok to be interested in money for its own sake; whatever floats your boat. just so long as you're conscious of your motivations. the thing to avoid is *unconsciously* letting the need for money warp your ideas about what kind of work you find most interesting.

9 many people face this question on a smaller scale with individual projects. but it's easier both to recognize and to accept a dead end in a single project than to abandon some type of work entirely. the more determined you are, the harder it gets. like a spanish flu victim, you're fighting your own immune system: instead of giving up, you tell yourself, i should just try harder. and who can say you're not right?

thanks to trevor blackwell, john carmack, john collison, patrick collison, robert morris, geoff ralston, and harj taggar for reading drafts of this.

successful people tend to be persistent. new ideas often don't work at first, but they're not deterred. they keep trying and eventually find something that does.

mere obstinacy, on the other hand, is a recipe for failure. obstinate people are so annoying. they won't listen. they beat their heads against a wall and get nowhere.

but is there any real difference between these two cases? are persistent and obstinate people actually behaving differently? or are they doing the same thing, and we just label them later as persistent or obstinate depending on whether they turned out to be right or not?

if that's the only difference then there's nothing to be learned from the distinction. telling someone to be persistent rather than obstinate would just be telling them to be right rather than wrong, and they already know that. whereas if persistence and obstinacy are actually different kinds of behavior, it would be worthwhile to tease them apart. [1]

i've talked to a lot of determined people, and it seems to me that they're different kinds of behavior. i've often walked away from a conversation thinking either "wow, that guy is determined" or "damn, that guy is stubborn," and i don't think i'm just talking about whether they seemed right or not. that's part of it, but not all of it.

there's something annoying about the obstinate that's not simply due to being mistaken. they won't listen. and that's not true of all determined people. i can't think of anyone more determined than the collison brothers, and when you point out a problem to them, they not only listen, but listen with an almost predatory intensity. is there a hole in the bottom of their boat? probably not, but if there is, they want to know about it.

it's the same with most successful people. they're never *more* engaged than when you disagree with them. whereas the obstinate don't want to hear you. when you point out problems, their eyes glaze over, and their replies sound like ideologues talking about matters of doctrine. [2]

the reason the persistent and the obstinate seem similar is that they're both hard to stop. but they're hard to stop in different senses. the persistent are like boats whose engines can't be throttled back. the obstinate are like boats whose rudders can't be turned. [3]

in the degenerate case they're indistinguishable: when there's only one way to solve a problem, your only choice is whether to give up or not, and persistence and obstinacy both say no. this is presumably why the two are so often conflated in popular culture. it assumes simple problems. but as problems get more complicated, we can see the difference between them. the persistent are much more attached to points high in the decision tree than to minor ones lower down, while the obstinate spray "don't give up" indiscriminately over the whole tree.

the persistent are attached to the goal. the obstinate are attached to their ideas about how to reach it.

worse still, that means they'll tend to be attached to their *first* ideas about how to solve a problem, even though these are the least informed by the experience of working on it. so the obstinate aren't merely attached to details, but disproportionately likely to be attached to wrong ones.

why are they like this? why are the obstinate obstinate? one possibility is that they're overwhelmed. they're not very capable. they take on a hard problem. they're immediately in over their head. so they grab onto ideas the way someone on the deck of a rolling ship might grab onto the nearest handhold.

that was my initial theory, but on examination it doesn't hold up. if being obstinate were simply a consequence of being in over one's head, you could make persistent people become obstinate by making

empirically this doesn't seem to be a problem. it's ok to be irrational near the top of the decision tree. one reason is that we humans will work harder on a problem we love. but there's another more subtle factor involved as well: our preferences among problems aren't random. when we love a problem that other people don't, it's often because we've unconsciously noticed that it's more important than they realize.

which leads to our fifth quality: there needs to be some overall goal. if you're like me you began, as a kid, merely with the desire to do something great. in theory that should be the most powerful motivator of all, since it includes everything that could possibly be done. but in practice it's not much use, precisely because it includes too much. it doesn't tell you what to do at this moment.

so in practice your energy and imagination and resilience and good judgement have to be directed toward some fairly specific goal. not too specific, or you might miss a great discovery adjacent to what you're searching for, but not too general, or it won't work to motivate you. [6]

when you look at the internal structure of persistence, it doesn't resemble obstinacy at all. it's so much more complex. five distinct qualities — energy, imagination, resilience, good judgement, and focus on a goal — combine to produce a phenomenon that seems a bit like obstinacy in the sense that it causes you not to give up. but the way you don't give up is completely different. instead of merely resisting change, you're driven toward a goal by energy and resilience, through paths discovered by imagination and optimized by judgement. you'll give way on any point low down in the decision tree, if its expected value drops sufficiently, but energy and resilience keep pushing you toward whatever you chose higher up.

considering what it's made of, it's not surprising that the right kind of stubbornness is so much rarer than the wrong kind, or that it gets so much better results. anyone can do obstinacy. indeed, kids and drunks and fools are best at it. whereas very few people have enough of all five of the qualities that produce the right kind of stubbornness, but when they do the results are magical.

---

notes

1 i'm going to use "persistent" for the good kind of stubborn and "obstinate" for the bad kind, but i can't claim i'm simply following current usage. conventional opinion barely distinguishes between good and bad kinds of stubbornness, and usage is correspondingly promiscuous. i could have invented a new word for the good kind, but it seemed better just to stretch "persistent."

2 there are some domains where one can succeed by being obstinate. some political leaders have been notorious for it. but it won't work in situations where you have to pass external tests. and indeed the political leaders who are famous for being obstinate are famous for getting power, not for using it well.

3 there will be some resistance to turning the rudder of a persistent person, because there's some cost to changing direction.

4 the obstinate do sometimes succeed in solving hard problems. one way is through luck: like the stopped clock that's right twice a day, they seize onto some arbitrary idea, and it turns out to be right. another is when their obstinacy cancels out some other form of error. for example, if a leader has overcautious subordinates, their estimates of the probability of success will always be off in the same direction. so if he mindlessly says "push ahead regardless" in every borderline case, he'll usually turn out to be right.

5 if you stop there, at just energy and imagination, you get the conventional caricature of an artist or poet.

what should one do? that may seem a strange question, but it's not meaningless or unanswerable. it's the sort of question kids ask before they learn not to ask big questions. i only came across it myself in the process of investigating something else. but once i did, i thought i should at least try to answer it.

so what *should* one do? one should help people, and take care of the world. those two are obvious. but is there anything else? when i ask that, the answer that pops up is *make good new things*.

i can't prove that one should do this, any more than i can prove that one should help people or take care of the world. we're talking about first principles here. but i can explain why this principle makes sense. the most impressive thing humans can do is to think. it may be the most impressive thing that can be done. and the best kind of thinking, or more precisely the best proof that one has thought well, is to make good new things.

i mean new things in a very general sense. newton's physics was a good new thing. indeed, the first version of this principle was to have good new ideas. but that didn't seem general enough: it didn't include making art or music, for example, except insofar as they embody new ideas. and while they may embody new ideas, that's not all they embody, unless you stretch the word "idea" so uselessly thin that it includes everything that goes through your nervous system.

even for ideas that one has consciously, though, i prefer the phrasing "make good new things." there are other ways to describe the best kind of thinking. to make discoveries, for example, or to understand something more deeply than others have. but how well do you understand something if you can't make a model of it, or write about it? indeed, trying to express what you understand is not just a way to prove that you understand it, but a way to understand it better.

another reason i like this phrasing is that it biases us toward creation. it causes us to prefer the kind of ideas that are naturally seen as making things rather than, say, making critical observations about things other people have made. those are ideas too, and sometimes valuable ones, but it's easy to trick oneself into believing they're more valuable than they are. criticism seems sophisticated, and making new things often seems awkward, especially at first; and yet it's precisely those first steps that are most rare and valuable.

is newness essential? i think so. obviously it's essential in science. if you copied a paper of someone else's and published it as your own, it would seem not merely unimpressive but dishonest. and it's similar in the arts. a copy of a good painting can be a pleasing thing, but it's not impressive in the way the original was. which in turn implies it's not impressive to make the same thing over and over, however well; you're just copying yourself.

note though that we're talking about a different kind of should with this principle. taking care of people and the world are shoulds in the sense that they're one's duty, but making good new things is a should in the sense that this is how to live to one's full potential. historically most rules about how to live have been a mix of both kinds of should, though usually with more of the former than the latter. [1]

for most of history the question "what should one do?" got much the same answer everywhere, whether you asked cicero or confucius. you should be wise, brave, honest, temperate, and just, uphold tradition, and serve the public interest. there was a long stretch where in some parts of the world the answer became "serve god," but in practice it was still considered good to be wise, brave, honest, temperate, and just, uphold tradition, and serve the public interest. and indeed this recipe would have seemed right to most victorians. but there's nothing in it about taking care of the world or making new things, and that's a bit worrying, because it seems like this question should be a timeless one. the answer shouldn't change much.

notes

1 we could treat all three as the same kind of should by saying that it's one's duty to live well — for example by saying, as some christians have, that it's one's duty to make the most of one's god-given gifts. but this seems one of those casuistries people invented to evade the stern requirements of religion: it was permissible to spend time studying math instead of praying or performing acts of charity because otherwise you were rejecting a gift god had given you. a useful casuistry no doubt, but we don't need it.

we could also combine the first two principles, since people are part of the world. why should our species get special treatment? i won't try to justify this choice, but i'm skeptical that anyone who claims to think differently actually lives according to their principles.

2 confucius was also excluded from public life after ending up on the losing end of a power struggle, and presumably he too would not be so famous now if it hadn't been for this long stretch of enforced leisure.

thanks to trevor blackwell, jessica livingston, and robert morris for reading drafts of this.

there's some debate about whether it's a good idea to "follow your passion." in fact the question is impossible to answer with a simple yes or no. sometimes you should and sometimes you shouldn't, but the border between should and shouldn't is very complicated. the only way to give a general answer is to trace it.

when people talk about this question, there's always an implicit "instead of." all other things being equal, why wouldn't you work on what interests you the most? so even raising the question implies that all other things aren't equal, and that you have to choose between working on what interests you the most and something else, like what pays the best.

and indeed if your main goal is to make money, you can't usually afford to work on what interests you the most. people pay you for doing what they want, not what you want. but there's an obvious exception: when you both want the same thing. for example, if you love football, and you're good enough at it, you can get paid a lot to play it.

of course the odds are against you in a case like football, because so many other people like playing it too. this is not to say you shouldn't try though. it depends how much ability you have and how hard you're willing to work.

the odds are better when you have strange tastes: when you like something that pays well and that few other people like. for example, it's clear that bill gates truly loved running a software company. he didn't just love programming, which a lot of people do. he loved writing software for customers. that is a very strange taste indeed, but if you have it, you can make a lot by indulging it.

there are even some people who have a genuine intellectual interest in making money. this is distinct from mere greed. they just can't help noticing when something is mispriced, and can't help doing something about it. it's like a puzzle for them. [1]

in fact there's an edge case here so spectacular that it turns all the preceding advice on its head. if you want to make a really huge amount of money — hundreds of millions or even billions of dollars — it turns out to be very useful to work on what interests you the most. the reason is not the extra motivation you get from doing this, but that the way to make a really large amount of money is to start a startup, and working on what interests you is an excellent way to discover startup ideas.

many if not most of the biggest startups began as projects the founders were doing for fun. apple, google, and facebook all began that way. why is this pattern so common? because the best ideas tend to be such outliers that you'd overlook them if you were consciously looking for ways to make money. whereas if you're young and good at technology, your unconscious instincts about what would be interesting to work on are very well aligned with what needs to be built.

so there's something like a midwit peak for making money. if you don't need to make much, you can work on whatever you're most interested in; if you want to become moderately rich, you can't usually afford to; but if you want to become super rich, and you're young and good at technology, working on what you're most interested in becomes a good idea again.

what if you're not sure what you want? what if you're attracted to the idea of making money and more attracted to some kinds of work than others, but neither attraction predominates? how do you break ties?

the key here is to understand that such ties are only apparent. when you have trouble choosing between following your interests and making money, it's never because you have complete knowledge of yourself and of the types of work you're choosing between, and the options are perfectly balanced.

notes

1 these examples show why it's a mistake to assume that economic inequality must be evidence of some kind of brokenness or unfairness. it's obvious that different people have different interests, and that some interests yield far more money than others, so how can it not be obvious that some people will end up much richer than others? in a world where some people like to write enterprise software and others like to make studio pottery, economic inequality is the natural outcome.

2 difficulty choosing between interests is a different matter. that's not always due to ignorance. it's often intrinsically difficult. i still have trouble doing it.

3 you can't always take people at their word on this. since it's more prestigious to work on things you're interested in than to be driven by money, people who are driven mainly by money will often claim to be more interested in their work than they actually are. one way to test such claims is by doing the following thought experiment: if their work didn't pay well, would they take day jobs doing something else in order to do it in their spare time? lots of mathematicians and scientists and engineers would. historically lots *have*. but i don't think as many investment bankers would.

this thought experiment is also useful for distinguishing between university departments.

thanks to trevor blackwell, paul buchheit, jessica livingston, robert morris, harj taggar, and garry tan for reading drafts of this.

life is short, as everyone knows. when i was a kid i used to wonder about this. is life actually short, or are we really complaining about its finiteness? would we be just as likely to feel life was short if we lived 10 times as long?

since there didn't seem any way to answer this question, i stopped wondering about it. then i had kids. that gave me a way to answer the question, and the answer is that life actually is short.

having kids showed me how to convert a continuous quantity, time, into discrete quantities. you only get 52 weekends with your 2 year old. if christmas-as-magic lasts from say ages 3 to 10, you only get to watch your child experience it 8 times. and while it's impossible to say what is a lot or a little of a continuous quantity like time, 8 is not a lot of something. if you had a handful of 8 peanuts, or a shelf of 8 books to choose from, the quantity would definitely seem limited, no matter what your lifespan was.

ok, so life actually is short. does it make any difference to know that?

it has for me. it means arguments of the form "life is too short for x" have great force. it's not just a figure of speech to say that life is too short for something. it's not just a synonym for annoying. if you find yourself thinking that life is too short for something, you should try to eliminate it if you can.

when i ask myself what i've found life is too short for, the word that pops into my head is "bullshit." i realize that answer is somewhat tautological. it's almost the definition of bullshit that it's the stuff that life is too short for. and yet bullshit does have a distinctive character. there's something fake about it. it's the junk food of experience. [1]

if you ask yourself what you spend your time on that's bullshit, you probably already know the answer. unnecessary meetings, pointless disputes, bureaucracy, posturing, dealing with other people's mistakes, traffic jams, addictive but unrewarding pastimes.

there are two ways this kind of thing gets into your life: it's either forced on you, or it tricks you. to some extent you have to put up with the bullshit forced on you by circumstances. you need to make money, and making money consists mostly of errands. indeed, the law of supply and demand ensures that: the more rewarding some kind of work is, the cheaper people will do it. it may be that less bullshit is forced on you than you think, though. there has always been a stream of people who opt out of the default grind and go live somewhere where opportunities are fewer in the conventional sense, but life feels more authentic. this could become more common.

you can do it on a smaller scale without moving. the amount of time you have to spend on bullshit varies between employers. most large organizations (and many small ones) are steeped in it. but if you consciously prioritize bullshit avoidance over other factors like money and prestige, you can probably find employers that will waste less of your time.

if you're a freelancer or a small company, you can do this at the level of individual customers. if you fire or avoid toxic customers, you can decrease the amount of bullshit in your life by more than you decrease your income.

but while some amount of bullshit is inevitably forced on you, the bullshit that sneaks into your life by tricking you is no one's fault but your own. and yet the bullshit you choose may be harder to eliminate than the bullshit that's forced on you. things that lure you into wasting your time have to be really good at tricking you. an example that will be familiar to a lot of people is arguing online. when someone contradicts you, they're in a sense attacking you. sometimes pretty overtly. your instinct when attacked is to defend yourself. but like a lot of instincts, this one wasn't designed for the world we now

you from pausing to savor life amid the daily slurry of errands and alarms. one of the most striking things i've read was not in a book, but the title of one: james salter's *burning the days*.

it is possible to slow time somewhat. i've gotten better at it. kids help. when you have small children, there are a lot of moments so perfect that you can't help noticing.

it does help too to feel that you've squeezed everything out of some experience. the reason i'm sad about my mother is not just that i miss her but that i think of all the things we could have done that we didn't. my oldest son will be 7 soon. and while i miss the 3 year old version of him, i at least don't have any regrets over what might have been. we had the best time a daddy and a 3 year old ever had.

relentlessly prune bullshit, don't wait to do things that matter, and savor the time you have. that's what you do when life is short.

---

notes

1 at first i didn't like it that the word that came to mind was one that had other meanings. but then i realized the other meanings are fairly closely related. bullshit in the sense of things you waste your time on is a lot like intellectual bullshit.

2 i chose this example deliberately as a note to self. i get attacked a lot online. people tell the craziest lies about me. and i have so far done a pretty mediocre job of suppressing the natural human inclination to say "hey, that's not true!"

thanks to jessica livingston and geoff ralston for reading drafts of this.

if you collected lists of techniques for doing great work in a lot of different fields, what would the intersection look like? i decided to find out by making it.

partly my goal was to create a guide that could be used by someone working in any field. but i was also curious about the shape of the intersection. and one thing this exercise shows is that it does have a definite shape; it's not just a point labelled "work hard."

the following recipe assumes you're very ambitious.

the first step is to decide what to work on. the work you choose needs to have three qualities: it has to be something you have a natural aptitude for, that you have a deep interest in, and that offers scope to do great work.

in practice you don't have to worry much about the third criterion. ambitious people are if anything already too conservative about it. so all you need to do is find something you have an aptitude for and great interest in. [1]

that sounds straightforward, but it's often quite difficult. when you're young you don't know what you're good at or what different kinds of work are like. some kinds of work you end up doing may not even exist yet. so while some people know what they want to do at 14, most have to figure it out.

the way to figure out what to work on is by working. if you're not sure what to work on, guess. but pick something and get going. you'll probably guess wrong some of the time, but that's fine. it's good to know about multiple things; some of the biggest discoveries come from noticing connections between different fields.

develop a habit of working on your own projects. don't let "work" mean something other people tell you to do. if you do manage to do great work one day, it will probably be on a project of your own. it may be within some bigger project, but you'll be driving your part of it.

what should your projects be? whatever seems to you excitingly ambitious. as you grow older and your taste in projects evolves, exciting and important will converge. at 7 it may seem excitingly ambitious to build huge things out of lego, then at 14 to teach yourself calculus, till at 21 you're starting to explore unanswered questions in physics. but always preserve excitingness.

there's a kind of excited curiosity that's both the engine and the rudder of great work. it will not only drive you, but if you let it have its way, will also show you what to work on.

what are you excessively curious about — curious to a degree that would bore most other people? that's what you're looking for.

once you've found something you're excessively interested in, the next step is to learn enough about it to get you to one of the frontiers of knowledge. knowledge expands fractally, and from a distance its edges look smooth, but once you learn enough to get close to one, they turn out to be full of gaps.

the next step is to notice them. this takes some skill, because your brain wants to ignore such gaps in order to make a simpler model of the world. many discoveries have come from asking questions about things that everyone else took for granted. [2]

if the answers seem strange, so much the better. great work often has a tincture of strangeness. you see this from painting to math. it would be affected to try to manufacture it, but if it appears, embrace it.

boldly chase outlier ideas, even if other people aren't interested in them — in fact, especially if they aren't. if you're excited about some possibility that everyone else ignores, and you have enough expert-

one sign that you're suited for some kind of work is when you like even the parts that other people find tedious or frightening.

but fields aren't people; you don't owe them any loyalty. if in the course of working on one thing you discover another that's more exciting, don't be afraid to switch.

if you're making something for people, make sure it's something they actually want. the best way to do this is to make something you yourself want. write the story you want to read; build the tool you want to use. since your friends probably have similar interests, this will also get you your initial audience.

this *should* follow from the excitingness rule. obviously the most exciting story to write will be the one you want to read. the reason i mention this case explicitly is that so many people get it wrong. instead of making what they want, they try to make what some imaginary, more sophisticated audience wants. and once you go down that route, you're lost. [6]

there are a lot of forces that will lead you astray when you're trying to figure out what to work on. pretentiousness, fashion, fear, money, politics, other people's wishes, eminent frauds. but if you stick to what you find genuinely interesting, you'll be proof against all of them. if you're interested, you're not astray.

following your interests may sound like a rather passive strategy, but in practice it usually means following them past all sorts of obstacles. you usually have to risk rejection and failure. so it does take a good deal of boldness.

but while you need boldness, you don't usually need much planning. in most cases the recipe for doing great work is simply: work hard on excitingly ambitious projects, and something good will come of it. instead of making a plan and then executing it, you just try to preserve certain invariants.

the trouble with planning is that it only works for achievements you can describe in advance. you can win a gold medal or get rich by deciding to as a child and then tenaciously pursuing that goal, but you can't discover natural selection that way.

i think for most people who want to do great work, the right strategy is not to plan too much. at each stage do whatever seems most interesting and gives you the best options for the future. i call this approach "staying upwind." this is how most people who've done great work seem to have done it.

even when you've found something exciting to work on, working on it is not always straightforward. there will be times when some new idea makes you leap out of bed in the morning and get straight to work. but there will also be plenty of times when things aren't like that.

you don't just put out your sail and get blown forward by inspiration. there are headwinds and currents and hidden shoals. so there's a technique to working, just as there is to sailing.

for example, while you must work hard, it's possible to work too hard, and if you do that you'll find you get diminishing returns: fatigue will make you stupid, and eventually even damage your health. the point at which work yields diminishing returns depends on the type. some of the hardest types you might only be able to do for four or five hours a day.

ideally those hours will be contiguous. to the extent you can, try to arrange your life so you have big blocks of time to work in. you'll shy away from hard tasks if you know you might be interrupted.

it will probably be harder to start working than to keep working. you'll often have to trick yourself to get over that initial threshold. don't worry about this; it's the nature of work, not a flaw in your character. work has a sort of activation energy, both per day and per project. and since this threshold is fake

the trouble with exponential growth is that the curve feels flat in the beginning. it isn't; it's still a wonderful exponential curve. but we can't grasp that intuitively, so we underrate exponential growth in its early stages.

something that grows exponentially can become so valuable that it's worth making an extraordinary effort to get it started. but since we underrate exponential growth early on, this too is mostly done unconsciously: people push through the initial, unrewarding phase of learning something new because they know from experience that learning new things always takes an initial push, or they grow their audience one fan at a time because they have nothing better to do. if people consciously realized they could invest in exponential growth, many more would do it.

work doesn't just happen when you're trying to. there's a kind of undirected thinking you do when walking or taking a shower or lying in bed that can be very powerful. by letting your mind wander a little, you'll often solve problems you were unable to solve by frontal attack.

you have to be working hard in the normal way to benefit from this phenomenon, though. you can't just walk around daydreaming. the daydreaming has to be interleaved with deliberate work that feeds it questions. [10]

everyone knows to avoid distractions at work, but it's also important to avoid them in the other half of the cycle. when you let your mind wander, it wanders to whatever you care about most at that moment. so avoid the kind of distraction that pushes your work out of the top spot, or you'll waste this valuable type of thinking on the distraction instead. (exception: don't avoid love.)

consciously cultivate your taste in the work done in your field. until you know which is the best and what makes it so, you don't know what you're aiming for.

and that *is* what you're aiming for, because if you don't try to be the best, you won't even be good. this observation has been made by so many people in so many different fields that it might be worth thinking about why it's true. it could be because ambition is a phenomenon where almost all the error is in one direction — where almost all the shells that miss the target miss by falling short. or it could be because ambition to be the best is a qualitatively different thing from ambition to be good. or maybe being good is simply too vague a standard. probably all three are true. [11]

fortunately there's a kind of economy of scale here. though it might seem like you'd be taking on a heavy burden by trying to be the best, in practice you often end up net ahead. it's exciting, and also strangely liberating. it simplifies things. in some ways it's easier to try to be the best than to try merely to be good.

one way to aim high is to try to make something that people will care about in a hundred years. not because their opinions matter more than your contemporaries', but because something that still seems good in a hundred years is more likely to be genuinely good.

don't try to work in a distinctive style. just try to do the best job you can; you won't be able to help doing it in a distinctive way.

style is doing things in a distinctive way without trying to. trying to is affectation.

affectation is in effect to pretend that someone other than you is doing the work. you adopt an impressive but fake persona, and while you're pleased with the impressiveness, the fakeness is what shows in the work. [12]

indeed, in some kinds of work it's good to strip whatever you're doing to its essence. the result will be more concentrated; you'll understand it better; and you won't be able to lie to yourself about whether there's anything real there.

mathematical elegance may sound like a mere metaphor, drawn from the arts. that's what i thought when i first heard the term "elegant" applied to a proof. but now i suspect it's conceptually prior — that the main ingredient in artistic elegance is mathematical elegance. at any rate it's a useful standard well beyond math.

elegance can be a long-term bet, though. laborious solutions will often have more prestige in the short term. they cost a lot of effort and they're hard to understand, both of which impress people, at least temporarily.

whereas some of the very best work will seem like it took comparatively little effort, because it was in a sense already there. it didn't have to be built, just seen. it's a very good sign when it's hard to say whether you're creating something or discovering it.

when you're doing work that could be seen as either creation or discovery, err on the side of discovery. try thinking of yourself as a mere conduit through which the ideas take their natural shape.

(strangely enough, one exception is the problem of choosing a problem to work on. this is usually seen as search, but in the best case it's more like creating something. in the best case you create the field in the process of exploring it.)

similarly, if you're trying to build a powerful tool, make it gratuitously unrestrictive. a powerful tool almost by definition will be used in ways you didn't expect, so err on the side of eliminating restrictions, even if you don't know what the benefit will be.

great work will often be tool-like in the sense of being something others build on. so it's a good sign if you're creating ideas that others could use, or exposing questions that others could answer. the best ideas have implications in many different areas.

if you express your ideas in the most general form, they'll be truer than you intended.

true by itself is not enough, of course. great ideas have to be true and new. and it takes a certain amount of ability to see new ideas even once you've learned enough to get to one of the frontiers of knowledge.

in english we give this ability names like originality, creativity, and imagination. and it seems reasonable to give it a separate name, because it does seem to some extent a separate skill. it's possible to have a great deal of ability in other respects — to have a great deal of what's often called *technical* ability — and yet not have much of this.

i've never liked the term "creative process." it seems misleading. originality isn't a process, but a habit of mind. original thinkers throw off new ideas about whatever they focus on, like an angle grinder throwing off sparks. they can't help it.

if the thing they're focused on is something they don't understand very well, these new ideas might not be good. one of the most original thinkers i know decided to focus on dating after he got divorced. he knew roughly as much about dating as the average 15 year old, and the results were spectacularly colorful. but to see originality separated from expertise like that made its nature all the more clear.

i don't know if it's possible to cultivate originality, but there are definitely ways to make the most of however much you have. for example, you're much more likely to have original ideas when you're

brought with them. but they didn't at the time; it took the greater part of a century for the heliocentric model to be generally accepted, even among astronomers, because it felt so wrong.

indeed, if you think about it, a good new idea has to seem bad to most people, or someone would have already explored it. so what you're looking for is ideas that seem crazy, but the right kind of crazy. how do you recognize these? you can't with certainty. often ideas that seem bad are bad. but ideas that are the right kind of crazy tend to be exciting; they're rich in implications; whereas ideas that are merely bad tend to be depressing.

there are two ways to be comfortable breaking rules: to enjoy breaking them, and to be indifferent to them. i call these two cases being aggressively and passively independent-minded.

the aggressively independent-minded are the naughty ones. rules don't merely fail to stop them; breaking rules gives them additional energy. for this sort of person, delight at the sheer audacity of a project sometimes supplies enough activation energy to get it started.

the other way to break rules is not to care about them, or perhaps even to know they exist. this is why novices and outsiders often make new discoveries; their ignorance of a field's assumptions acts as a source of temporary passive independent-mindedness. aspies also seem to have a kind of immunity to conventional beliefs. several i know say that this helps them to have new ideas.

strictness plus rule-breaking sounds like a strange combination. in popular culture they're opposed. but popular culture has a broken model in this respect. it implicitly assumes that issues are trivial ones, and in trivial matters strictness and rule-breaking *are* opposed. but in questions that really matter, only rule-breakers can be truly strict.

an overlooked idea often doesn't lose till the semifinals. you do see it, subconsciously, but then another part of your subconscious shoots it down because it would be too weird, too risky, too much work, too controversial. this suggests an exciting possibility: if you could turn off such filters, you could see more new ideas.

one way to do that is to ask what would be good ideas for *someone else* to explore. then your subconscious won't shoot them down to protect you.

you could also discover overlooked ideas by working in the other direction: by starting from what's obscuring them. every cherished but mistaken principle is surrounded by a dead zone of valuable ideas that are unexplored because they contradict it.

religions are collections of cherished but mistaken principles. so anything that can be described either literally or metaphorically as a religion will have valuable unexplored ideas in its shadow. copernicus and darwin both made discoveries of this type. [18]

what are people in your field religious about, in the sense of being too attached to some principle that might not be as self-evident as they think? what becomes possible if you discard it?

people show much more originality in solving problems than in deciding which problems to solve. even the smartest can be surprisingly conservative when deciding what to work on. people who'd never dream of being fashionable in any other way get sucked into working on fashionable problems.

one reason people are more conservative when choosing problems than solutions is that problems are bigger bets. a problem could occupy you for years, while exploring a solution might only take days. but even so i think most people are too conservative. they're not merely responding to risk, but to fashion as well. unfashionable problems are undervalued.

it's a great thing to be rich in unanswered questions. and this is one of those situations where the rich get richer, because the best way to acquire new questions is to try answering existing ones. questions don't just lead to answers, but also to more questions.

the best questions grow in the answering. you notice a thread protruding from the current paradigm and try pulling on it, and it just gets longer and longer. so don't require a question to be obviously big before you try answering it. you can rarely predict that. it's hard enough even to notice the thread, let alone to predict how much will unravel if you pull on it.

it's better to be promiscuously curious — to pull a little bit on a lot of threads, and see what happens. big things start small. the initial versions of big things were often just experiments, or side projects, or talks, which then grew into something bigger. so start lots of small things.

being prolific is underrated. the more different things you try, the greater the chance of discovering something new. understand, though, that trying lots of things will mean trying lots of things that don't work. you can't have a lot of good ideas without also having a lot of bad ones. [21]

though it sounds more responsible to begin by studying everything that's been done before, you'll learn faster and have more fun by trying stuff. and you'll understand previous work better when you do look at it. so err on the side of starting. which is easier when starting means starting small; those two ideas fit together like two puzzle pieces.

how do you get from starting small to doing something great? by making successive versions. great things are almost always made in successive versions. you start with something small and evolve it, and the final version is both cleverer and more ambitious than anything you could have planned.

it's particularly useful to make successive versions when you're making something for people — to get an initial version in front of them quickly, and then evolve it based on their response.

begin by trying the simplest thing that could possibly work. surprisingly often, it does. if it doesn't, this will at least get you started.

don't try to cram too much new stuff into any one version. there are names for doing this with the first version (taking too long to ship) and the second (the second system effect), but these are both merely instances of a more general principle.

an early version of a new project will sometimes be dismissed as a toy. it's a good sign when people do this. that means it has everything a new idea needs except scale, and that tends to follow. [22]

the alternative to starting with something small and evolving it is to plan in advance what you're going to do. and planning does usually seem the more responsible choice. it sounds more organized to say "we're going to do x and then y and then z" than "we're going to try x and see what happens." and it is more *organized*; it just doesn't work as well.

planning per se isn't good. it's sometimes necessary, but it's a necessary evil — a response to unforgiving conditions. it's something you have to do because you're working with inflexible media, or because you need to coordinate the efforts of a lot of people. if you keep projects small and use flexible media, you don't have to plan as much, and your designs can evolve instead.

take as much risk as you can afford. in an efficient market, risk is proportionate to reward, so don't look for certainty, but for a bet with high expected value. if you're not failing occasionally, you're probably being too conservative.

the sooner you overcome this passivity, the better. if you're still in school, try thinking of your education as your project, and your teachers as working for you rather than vice versa. that may seem a stretch, but it's not merely some weird thought experiment. it's the truth economically, and in the best case it's the truth intellectually as well. the best teachers don't want to be your bosses. they'd prefer it if you pushed ahead, using them as a source of advice, rather than being pulled by them through the material.

schools also give you a misleading impression of what work is like. in school they tell you what the problems are, and they're almost always soluble using no more than you've been taught so far. in real life you have to figure out what the problems are, and you often don't know if they're soluble at all.

but perhaps the worst thing schools do to you is train you to win by hacking the test. you can't do great work by doing that. you can't trick god. so stop looking for that kind of shortcut. the way to beat the system is to focus on problems and solutions that others have overlooked, not to skimp on the work itself.

don't think of yourself as dependent on some gatekeeper giving you a "big break." even if this were true, the best way to get it would be to focus on doing good work rather than chasing influential people.

and don't take rejection by committees to heart. the qualities that impress admissions officers and prize committees are quite different from those required to do great work. the decisions of selection committees are only meaningful to the extent that they're part of a feedback loop, and very few are.

people new to a field will often copy existing work. there's nothing inherently bad about that. there's no better way to learn how something works than by trying to reproduce it. nor does copying necessarily make your work unoriginal. originality is the presence of new ideas, not the absence of old ones.

there's a good way to copy and a bad way. if you're going to copy something, do it openly instead of furtively, or worse still, unconsciously. this is what's meant by the famously misattributed phrase "great artists steal." the really dangerous kind of copying, the kind that gives copying a bad name, is the kind that's done without realizing it, because you're nothing more than a train running on tracks laid down by someone else. but at the other extreme, copying can be a sign of superiority rather than subordination. [25]

in many fields it's almost inevitable that your early work will be in some sense based on other people's. projects rarely arise in a vacuum. they're usually a reaction to previous work. when you're first starting out, you don't have any previous work; if you're going to react to something, it has to be someone else's. once you're established, you can react to your own. but while the former gets called derivative and the latter doesn't, structurally the two cases are more similar than they seem.

oddly enough, the very novelty of the most novel ideas sometimes makes them seem at first to be more derivative than they are. new discoveries often have to be conceived initially as variations of existing things, *even by their discoverers*, because there isn't yet the conceptual vocabulary to express them.

there are definitely some dangers to copying, though. one is that you'll tend to copy old things — things that were in their day at the frontier of knowledge, but no longer are.

and when you do copy something, don't copy every feature of it. some will make you ridiculous if you do. don't copy the manner of an eminent 50 year old professor if you're 18, for example, or the idiom of a renaissance poem hundreds of years later.

husband your morale. it's the basis of everything when you're working on ambitious projects. you have to nurture and protect it like a living organism.

morale starts with your view of life. you're more likely to do great work if you're an optimist, and more likely to if you think of yourself as lucky than if you think of yourself as a victim.

indeed, work can to some extent protect you from your problems. if you choose work that's pure, its very difficulties will serve as a refuge from the difficulties of everyday life. if this is escapism, it's a very productive form of it, and one that has been used by some of the greatest minds in history.

morale compounds via work: high morale helps you do good work, which increases your morale and helps you do even better work. but this cycle also operates in the other direction: if you're not doing good work, that can demoralize you and make it even harder to. since it matters so much for this cycle to be running in the right direction, it can be a good idea to switch to easier work when you're stuck, just so you start to get something done.

one of the biggest mistakes ambitious people make is to allow setbacks to destroy their morale all at once, like a balloon bursting. you can inoculate yourself against this by explicitly considering setbacks a part of your process. solving hard problems always involves some backtracking.

doing great work is a depth-first search whose root node is the desire to. so "if at first you don't succeed, try, try again" isn't quite right. it should be: if at first you don't succeed, either try again, or backtrack and then try again.

"never give up" is also not quite right. obviously there are times when it's the right choice to eject. a more precise version would be: never let setbacks panic you into backtracking more than you need to. corollary: never abandon the root node.

it's not necessarily a bad sign if work is a struggle, any more than it's a bad sign to be out of breath while running. it depends how fast you're running. so learn to distinguish good pain from bad. good pain is a sign of effort; bad pain is a sign of damage.

an audience is a critical component of morale. if you're a scholar, your audience may be your peers; in the arts, it may be an audience in the traditional sense. either way it doesn't need to be big. the value of an audience doesn't grow anything like linearly with its size. which is bad news if you're famous, but good news if you're just starting out, because it means a small but dedicated audience can be enough to sustain you. if a handful of people genuinely love what you're doing, that's enough.

to the extent you can, avoid letting intermediaries come between you and your audience. in some types of work this is inevitable, but it's so liberating to escape it that you might be better off switching to an adjacent type if that will let you go direct. [28]

the people you spend time with will also have a big effect on your morale. you'll find there are some who increase your energy and others who decrease it, and the effect someone has is not always what you'd expect. seek out the people who increase your energy and avoid those who decrease it. though of course if there's someone you need to take care of, that takes precedence.

don't marry someone who doesn't understand that you need to work, or sees your work as competition for your attention. if you're ambitious, you need to work; it's almost like a medical condition; so someone who won't let you work either doesn't understand you, or does and doesn't care.

ultimately morale is physical. you think with your body, so it's important to take care of it. that means exercising regularly, eating and sleeping well, and avoiding the more dangerous kinds of drugs. runn-

you're interested.

don't worry about being presumptuous. you don't have to tell anyone. and if it's too hard and you fail, so what? lots of people have worse problems than that. in fact you'll be lucky if it's the worst problem you have.

yes, you'll have to work hard. but again, lots of people have to work hard. and if you're working on something you find very interesting, which you necessarily will if you're on the right path, the work will probably feel less burdensome than a lot of your peers'.

the discoveries are out there, waiting to be made. why not by you?

---

notes

1 i don't think you could give a precise definition of what counts as great work. doing great work means doing something important so well that you expand people's ideas of what's possible. but there's no threshold for importance. it's a matter of degree, and often hard to judge at the time anyway. so i'd rather people focused on developing their interests rather than worrying about whether they're important or not. just try to do something amazing, and leave it to future generations to say if you succeeded.

2 a lot of standup comedy is based on noticing anomalies in everyday life. "did you ever notice...?" new ideas come from doing this about nontrivial things. which may help explain why people's reaction to a new idea is often the first half of laughing: ha!

3 that second qualifier is critical. if you're excited about something most authorities discount, but you can't give a more precise explanation than "they don't get it," then you're starting to drift into the territory of cranks.

4 finding something to work on is not simply a matter of finding a match between the current version of you and a list of known problems. you'll often have to coevolve with the problem. that's why it can sometimes be so hard to figure out what to work on. the search space is huge. it's the cartesian product of all possible types of work, both known and yet to be discovered, and all possible future versions of you.

there's no way you could search this whole space, so you have to rely on heuristics to generate promising paths through it and hope the best matches will be clustered. which they will not always be; different types of work have been collected together as much by accidents of history as by the intrinsic similarities between them.

5 there are many reasons curious people are more likely to do great work, but one of the more subtle is that, by casting a wide net, they're more likely to find the right thing to work on in the first place.

6 it can also be dangerous to make things for an audience you feel is less sophisticated than you, if that causes you to talk down to them. you can make a lot of money doing that, if you do it in a sufficiently cynical way, but it's not the route to great work. not that anyone using this m.o. would care.

7 this idea i learned from hardy's *a mathematician's apology*, which i recommend to anyone ambitious to do great work, in any field.

8 just as we overestimate what we can do in a day and underestimate what we can do over several years, we overestimate the damage done by procrastinating for a day and underestimate the damage done by procrastinating for several years.

9 you can't usually get paid for doing exactly what you want, especially early on. there are two options: get paid for doing work close to what you want and hope to push it closer, or get paid for doing something else entirely and do your own projects on the side. both can work, but both have drawbacks: in the first approach your work is compromised by default, and in the second you have to fight to get time to do it.

28 corollary: building something that enables people to go around intermediaries and engage directly with their audience is probably a good idea.

29 it may be helpful always to walk or run the same route, because that frees attention for thinking. it feels that way to me, and there is some historical evidence for it.

thanks to trevor blackwell, daniel gackle, pam graham, tom howard, patrick hsu, steve huffman, jessica livingston, henry lloyd-baker, bob metcalfe, ben miller, robert morris, michael nielsen, courtenay pipkin, joris poort, mieke roos, rajat suri, harj taggar, garry tan, and my younger son for suggestions and for reading drafts.